

Clinical Characteristics of Immunotherapy Data Cohorts

1. Baseline Patient and Disease Characteristics

Why We Are Doing This

What: We compare patient demographics, treatment histories, and survival outcomes across the 6 active immunotherapy trial cohorts.

Why: Before building predictive models or analysing transcriptomic signatures, we must understand the clinical composition of each dataset. Cohort-level differences in prior treatment, disease stage, and patient demographics can confound downstream survival and response analyses.

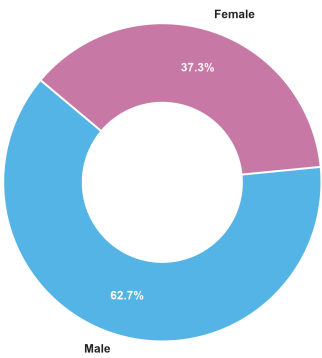
Question Answered: Are baseline patient populations sufficiently comparable across independent trial datasets to permit pooled multi-cohort machine learning?

This report compares patient demographics, treatments, survival, and sample attrition across the 6 active immunotherapy trial cohorts:

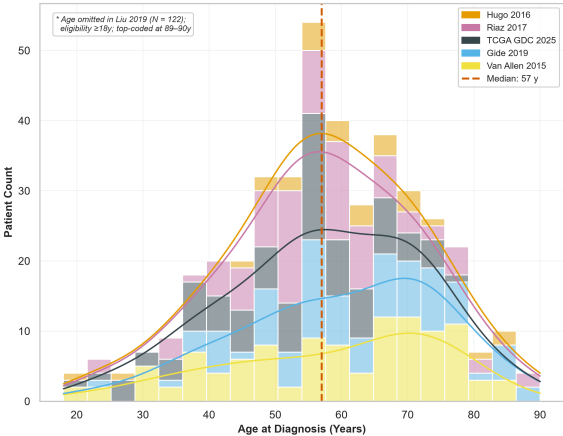
- **Liu 2019:** Immunotherapy trial cohort ($N = 122$).
- **Hugo 2016:** Immunotherapy trial cohort ($N = 27$).
- **Riaz 2017:** Immunotherapy trial cohort ($N = 107$).
- **TCGA GDC 2025:** Immunotherapy trial cohort ($N = 91$).
- **Gide 2019:** Immunotherapy trial cohort ($N = 91$).
- **Van Allen 2015:** Immunotherapy trial cohort ($N = 110$).

1.1 Clinical Demographics & Treatment Distributions

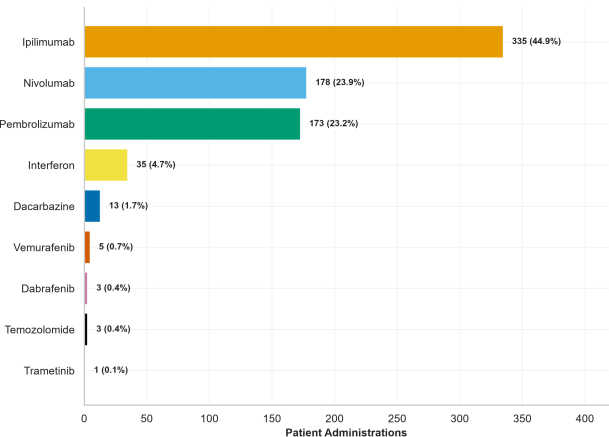
Clinical Patient Demographics & Treatment Distribution Across Immunotherapy Trial Cohorts



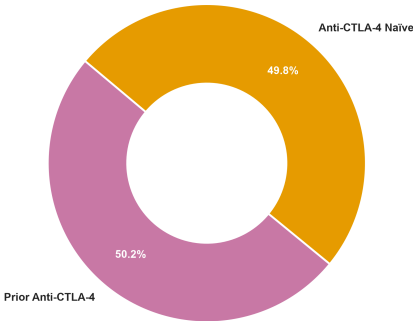
Panel A: Patient Sex Distribution ($N = 533$)



Panel B: Age Distribution (Median = 57, IQR = 48–68)



Panel C: Treatment Agents Administered ($N = 746$)



Panel D: Prior Anti-CTLA-4 Therapy Status ($N = 548$)

Figure 1: 2x2 Grid of Clinical Demographics and Treatment Histories across Immunotherapy Trial Cohorts. Panel A: sex distribution; Panel B: age at diagnosis; Panel C: treatment agents administered; Panel D: prior anti-CTLA-4 therapy status (Prior Ipilimumab vs Anti-CTLA-4 Naïve).

Key Demographics & Treatment Insights

- **Panel A: Sex Distribution** (*N* = 533): The overall trial cohort shows a male predominance (**62.7% Male** [*N* = 334] vs. **37.3% Female** [*N* = 199]), reflecting real-world cutaneous melanoma incidence patterns where male patients account for the majority of advanced presentations.
- **Panel B: Age Distribution across Studies:** Evaluated patient ages span from 18 to 90 years with a **median age of 57.0 years** (IQR: 48.0–68.0 years). Trial cohorts (Hugo 2016 : median 61.0; Riaz 2017 : median 56.0; TCGA GDC 2025 : median 55.0; Gide 2019 : median 61.0; Van Allen 2015 : median 61.5) display consistent age distributions centred around late middle age. *Note: Across annotated trial cohorts, ages range from 19 to 89 years (adult trial eligibility ≥ 18 years), with values top-coded/clipped at 89–90 years under HIPAA de-identification standards.*
- **Panel C: Treatment Agents Administered** (*N* = 746): Across all treatment administrations, the most frequent agents are **Pembrolizumab** (173 [23.2%]), **Nivolumab** (178 [23.9%]), **Ipilimumab** (335 [44.9%]), **Vemurafenib** (5 [0.7%]).
- **Panel D: Prior Anti-CTLA-4 Therapy Status** (*N* = 548): Across all trial patients, **50.2%** [*N* = 275] received prior anti-CTLA-4 therapy (Ipilimumab), while **49.8%** [*N* = 273] were anti-CTLA-4 naïve prior to anti-PD-1 initiation.

Table 1: Baseline Patient and Disease Characteristics

Characteristic	Liu 2019	Hugo 2016	Riaz 2017	TCGA GDC 2025	Gide 2019	Van Allen 2015	Total
N	122	27	107	91	91	110	548
Demographics							
Age, median (IQR)	N/A	61.0 (54.0-68.5)	56.0 (48.8-63.0)	55.0 (44.5-62.5)	61.0 (51.5-71.5)	61.5 (46.2-71.0)	57.0 (48.0-68.0)
Female sex, n (%)	51 (41.8%)	8 (29.6%)	47 (51.1%)	30 (33.0%)	31 (34.1%)	32 (29.1%)	199 (37.3%)
Treatment Agents & Exposure							
Agent — Pembrolizumab	71 (58.2%)	27 (100.0%)	0 (0.0%)	4 (4.4%)	71 (78.0%)	0 (0.0%)	173 (31.6%)
Agent — Nivolumab	51 (41.8%)	0 (0.0%)	107 (100.0%)	0 (0.0%)	20 (22.0%)	0 (0.0%)	178 (32.5%)
Agent — Ipilimumab	56 (45.9%)	0 (0.0%)	107 (100.0%)	21 (23.1%)	41 (45.1%)	110 (100.0%)	335 (61.1%)
Agent — Vemurafenib	0 (0.0%)	0 (0.0%)	0 (0.0%)	5 (5.5%)	0 (0.0%)	0 (0.0%)	5 (0.9%)
Agent — Dabrafenib	0 (0.0%)	0 (0.0%)	0 (0.0%)	3 (3.3%)	0 (0.0%)	0 (0.0%)	3 (0.5%)
Agent — Trametinib	0 (0.0%)	0 (0.0%)	0 (0.0%)	1 (1.1%)	0 (0.0%)	0 (0.0%)	1 (0.2%)
Agent — Dacarbazine	0 (0.0%)	0 (0.0%)	0 (0.0%)	13 (14.3%)	0 (0.0%)	0 (0.0%)	13 (2.4%)
Agent — Temozolomide	0 (0.0%)	0 (0.0%)	0 (0.0%)	3 (3.3%)	0 (0.0%)	0 (0.0%)	3 (0.5%)
Agent — Interferon	0 (0.0%)	0 (0.0%)	0 (0.0%)	35 (38.5%)	0 (0.0%)	0 (0.0%)	35 (6.4%)
Prior anti-CTLA-4 therapy	48 (39.3%)	0 (0.0%)	55 (51.4%)	21 (23.1%)	41 (45.1%)	110 (100.0%)	275 (50.2%)
Survival Outcomes							
Median OS, months (95% CI)	22.6	32.2	21.2	117.8	32.6	9.0	28.8
OS events, n (%)	62 (50.8%)	12 (46.2%)	63 (62.4%)	37 (41.1%)	36 (39.6%)	83 (75.5%)	293 (54.3%)
Median follow-up, months	17.5	14.4	17.8	54.2	20.5	9.1	20.6

2. Sample Preprocessing Attrition

Why We Are Doing This

What: We track sample retention through quality control, identifier standardisation, and clinical-expression alignment across all $N = 930$ initial records.

Why: Documenting attrition at each preprocessing step verifies data integrity and confirms that no patient sub-population was accidentally excluded.

Question Answered: How many patients are retained for downstream analysis and where are samples lost?

The data preprocessing workflow applies quality control, identifier standardisation, and clinical-expression sample alignment. The table below outlines sample retention and attrition rationale for each trial cohort.

Table 2: Sample Attrition Across Preprocessing Steps

Cohort	Preprocessing Step	N Initial	N Retained	N Removed	Rationale
Liu 2019	Clinical data loaded	122	122	0	Merged patient-level and sample-level clinical records.
Liu 2019	Clinical data harmonised	122	122	0	Applied identifier standardisation, clinical cleaning, baseline filtering, and variable harmonisation.
Liu 2019	Expression data availability	122	122	0	Identified samples with matching RNA-seq gene expression data.
Hugo 2016	Clinical data loaded	27	27	0	Merged patient-level and sample-level clinical records.
Hugo 2016	Clinical data harmonised	27	27	0	Applied identifier standardisation, clinical cleaning, baseline filtering, and variable harmonisation.
Hugo 2016	Expression data availability	27	27	0	Identified samples with matching RNA-seq gene expression data.
Riaz 2017	Clinical data loaded	107	107	0	Merged patient-level and sample-level clinical records.
Riaz 2017	Clinical data harmonised	107	107	0	Applied identifier standardisation, clinical cleaning, baseline filtering, and variable harmonisation.
Riaz 2017	Expression data availability	107	107	0	Identified samples with matching RNA-seq gene expression data.
TCGA GDC 2025	Clinical data loaded	473	473	0	Merged patient-level and sample-level clinical records.
TCGA GDC 2025	Clinical data harmonised	473	473	0	Applied identifier standardisation and clinical data cleaning.
TCGA GDC 2025	Expression data availability	473	472	1	Identified samples with matching RNA-seq gene expression data.
Gide 2019	Clinical data loaded	91	91	0	Merged patient-level and sample-level clinical records.
Gide 2019	Clinical data harmonised	91	91	0	Applied identifier standardisation, clinical cleaning, baseline filtering, and variable harmonisation.
Gide 2019	Expression data availability	91	91	0	Identified samples with matching RNA-seq gene expression data.
Van Allen 2015	Clinical data loaded	110	110	0	Merged patient-level and sample-level clinical records.
Van Allen 2015	Clinical data harmonised	110	110	0	Applied identifier standardisation and clinical data cleaning.
Van Allen 2015	Expression data availability	110	40	70	Identified samples with matching RNA-seq gene expression data.

Key Observations

- Overall Cohort Size ($N = 548$):** The largest individual dataset is **Liu 2019** ($N = 122$). Combined across all 6 trial cohorts, $N = 859$ cleaned patient records were harmonised for downstream analysis.
- Sample Attrition (92.4% Retention):** Across all 930 initial records, ****Liu 2019** retained 100% of samples ($N = 122$); **Hugo 2016** retained 100% of samples ($N = 27$); **Riaz 2017** retained 100% of samples ($N = 107$); **TCGA GDC 2025** lost 1 sample(s) ($473 \rightarrow 472$); **Gide 2019** retained 100% of samples ($N = 91$); **Van Allen 2015** lost 70 sample(s) ($110 \rightarrow 40$)**. All 6 immunotherapy trial cohorts retained 100% of their cleaned clinical and expression records.
- Follow-up Duration:** **TCGA GDC 2025** shows the longest median follow-up duration (**54.2 months**).
- Treatment History & Clinical Context:** Enrolled cohorts represent diverse treatment contexts including both treatment-naïve and pre-treated patient populations across Liu 2019, Hugo 2016, Riaz 2017, TCGA GDC 2025, Gide 2019, Van Allen 2015.

3. Overall Survival Curves (KM Plots)

Why We Are Doing This

What: We plot unstratified Kaplan-Meier overall survival curves for each of the 6 active immunotherapy trial cohorts.

Why: Visualising baseline mortality rates establishes the clinical context for each dataset and confirms that follow-up duration is sufficient to evaluate treatment outcomes.

Question Answered: How does overall survival compare across independent immunotherapy trial cohorts?

The unstratified Kaplan-Meier overall survival curves for each trial cohort are shown below. Median overall survival is annotated for each cohort where reached.

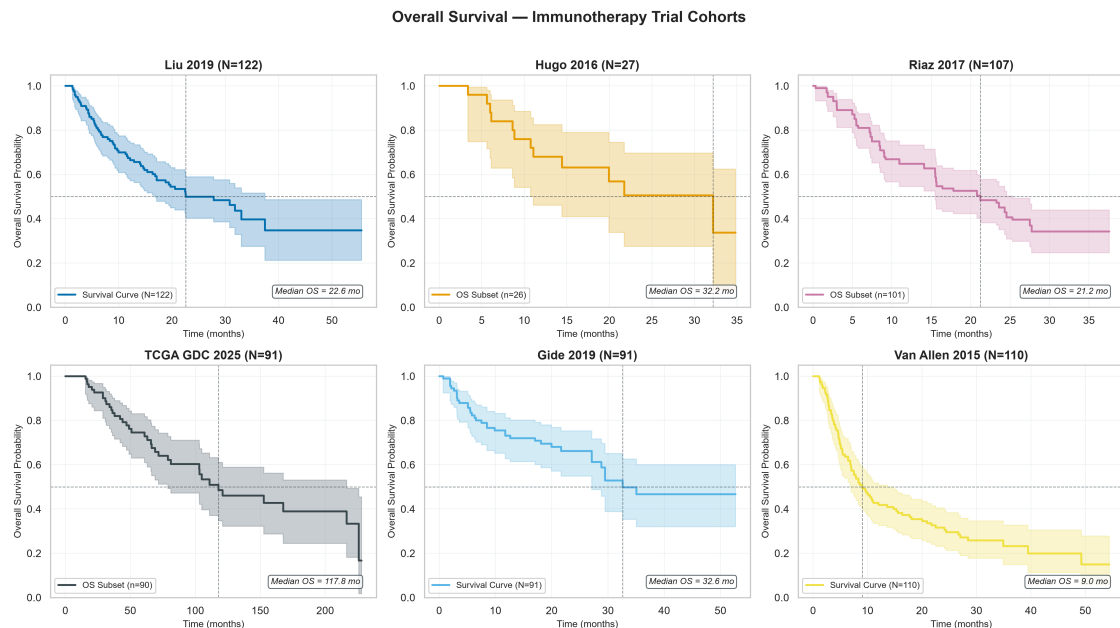


Figure 2: Unstratified Overall Survival KM Curves across All 6 Immunotherapy Trial Cohorts.

4. Overall Survival Stratified by Immunotherapy Response

Why We Are Doing This

What: We stratify Kaplan-Meier overall survival curves by RECIST clinical response status (Responders [CR/PR] vs. Non-responders [PD]) across the 6 immunotherapy trial cohorts ($N = 548$).

Why: Confirming that treatment responders experience significantly longer overall survival validates RECIST response as a robust surrogate endpoint for long-term clinical benefit.

Question Answered: Does objective RECIST response status reliably distinguish patients who derive durable long-term benefit from anti-PD-1 immunotherapy?

Overall Survival by Immunotherapy Response (RECIST)

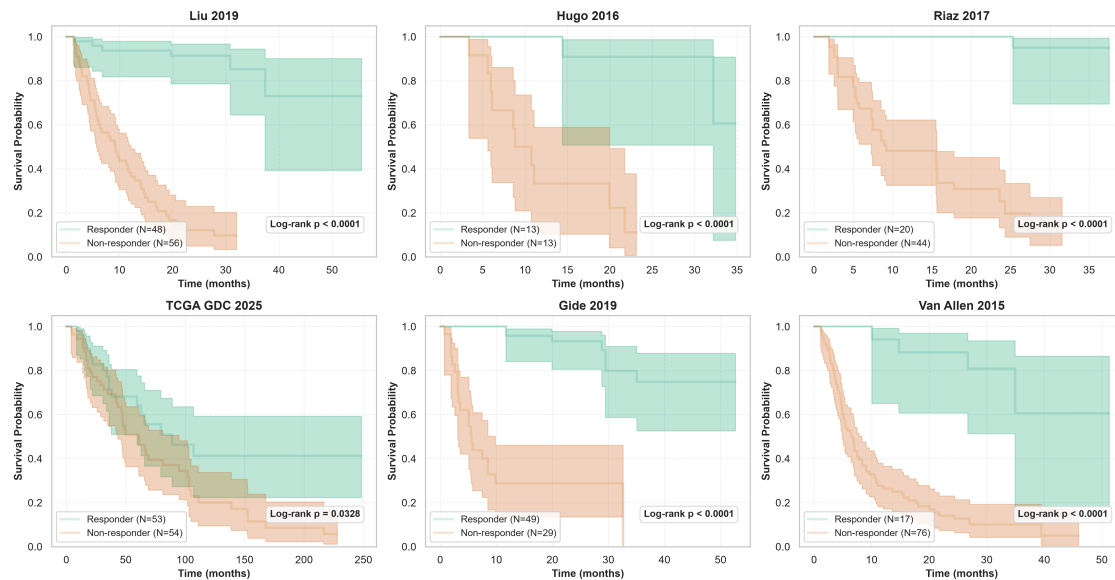


Figure 3: Overall Survival Stratified by RECIST Response Status across Immunotherapy Trial Cohorts. Treatment responders (CR/PR) exhibit significantly superior overall survival compared to non-responders (PD) across trial cohorts (Log-rank $p < 0.0001$).

🎯 Key Insights: Survival Stratification by Response

- 1. Profound Survival Benefit:** Treatment responders (CR/PR) achieve significantly longer overall survival compared to non-responders (PD) across independent trial cohorts (Liu 2019, Hugo 2016, Riaz 2017, TCGA GDC 2025, Gide 2019, Van Allen 2015; Log-rank $p < 0.0001$).
- 2. Durable Long-Term Survival:** Durable separation is consistently observed between responders and non-responders across Liu 2019, Hugo 2016, Riaz 2017, TCGA GDC 2025, Gide 2019, Van Allen 2015.
- 3. Conclusion:** Objective RECIST response is a highly robust surrogate endpoint for overall survival in metastatic melanoma, justifying its use as the primary outcome for predictive model training.

5. Technical Analysis Notes

⚠️ Methodological Limitations & Analytical Scope

- Unstratified Analysis:** All Kaplan-Meier curves are unstratified and descriptive. They do not adjust for demographic, clinical, molecular, treatment, or study-specific confounding factors.
- Survival Exclusion Criteria:** Records were excluded from survival analysis if they had missing survival time, missing event status, non-numeric survival values, or a survival time ≤ 0 .
- Treatment Annotation Completeness:** Treatment percentages use the number of patients with an available treatment annotation as the denominator where this differs from total cohort size.
- Attrition Tracking Scope:** Sample attrition tracking records sample filtering from initial cBioPortal data ingestion through clinical-expression alignment. Downstream feature engineering steps may impose additional filters not captured here.
- Median OS Reporting:** Where the estimated survival probability did not fall below 50% during follow-up, median OS is reported as **NR (not reached)**.



Clinical Analysis Script Execution & Software Module Architecture

- [run_clinical_analysis.py](#) : Generates baseline demographic grids, attrition metrics, cohort characteristics tables, Kaplan-Meier OS curves, and writes `cohort_characteristics_clinical.md`.
- [clean_data.py](#) : Preprocesses raw cohort clinical metadata and RNA-seq expression profiles into cleaned CSV matrices.
- [merge_datasets.py](#) : Merges processed expression matrices across cohorts into harmonised pooled matrices (`expr_merged.csv` , `clin_merged.csv`).
- [data_loaders.py](#) : Provides helper loader functions (`load_liu_2019` , `load_hugo_2016` , `load_riaz_2017`) for retrieving expression and clinical data.
- [styles.py](#) : Single source of truth for Okabe-Ito colour palettes (`COHORT_PALETTE` , `RESPONSE_PALETTE`) and visualisation presentation style.